

Knowledge	/10	Inquiry	15.5	/16	Application	/6
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Inquiry (16 Marks)

1. In a calorimetry experiment, a 2.45 g sample of magnesium is added to 250 mL of water. The temperature rose from 20.0°C to 26.5°C. Assume the density of water is 1.00 g/mL and that its specific heat capacity is 4.18 J/g°C. Calculate the molar enthalpy for the reaction in kJ/mol Mg. Show all conversions and equations used. USE GRASP (6 marks)

$$G: m_{Mg} = 2.45g \quad m_{H_2O} = 250mL \cdot \frac{1.00g}{mL} = 250g$$

$$R: \Delta H_r \text{ of Mg}$$

$$A: \Delta H = q_{sys} = -q_{sur} = -mc\Delta T = n\Delta H_r \quad n = \frac{m}{M}$$

$$S: n_{Mg} = \frac{2.45g}{24.31g/mol} = 0.1008 \text{ mol}$$

$$q_{sur} = mc\Delta T$$

$$= (250g)(4.18J/g^\circ C)(6.5^\circ C)$$

$$= 6792.5 J = \frac{1kJ}{1000J} = 6.7925 kJ$$

$$\Delta H = -6.7925 kJ$$

$$\Delta T_{H_2O} = 26.5^\circ C - 20.0^\circ C = 6.5^\circ C \quad c = 4.18 J/g^\circ C$$

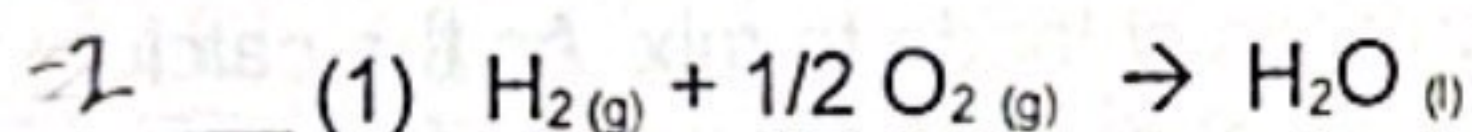
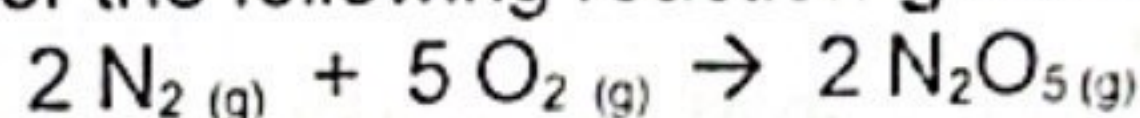
$$\Delta H_r = \frac{\Delta H}{n} = \frac{-6.7925 kJ}{0.1008 \text{ mol}}$$

$$= -67.398 kJ/mol$$

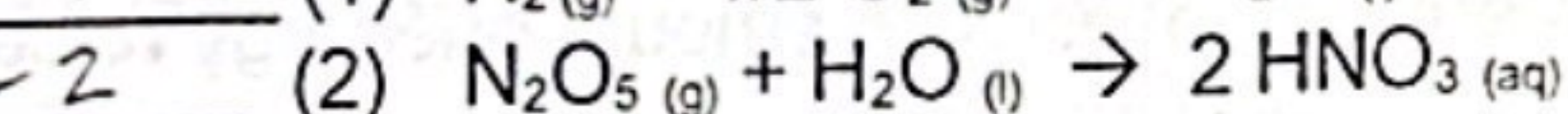
$$2 \text{ sig figs} = -67 kJ/mol$$

P. the molar enthalpy for the reaction is -67 kJ/mol

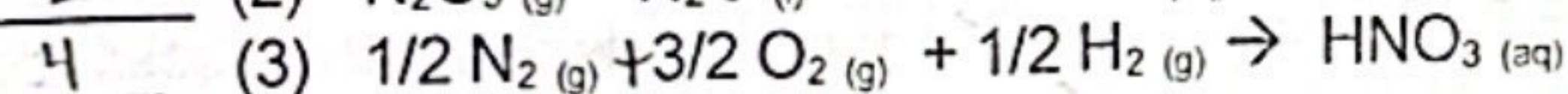
2. Determine the enthalpy change for the following reaction given the numbered reactions below: (6 marks)



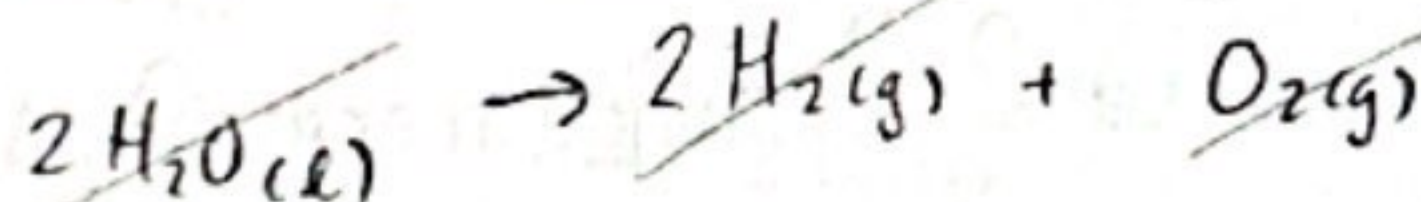
$$\Delta H_1 = -285.8 kJ$$



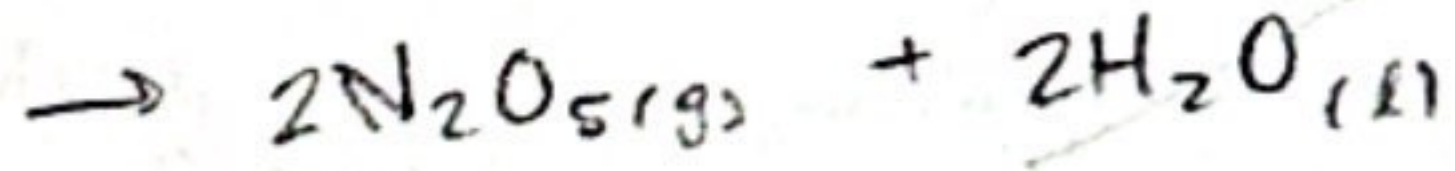
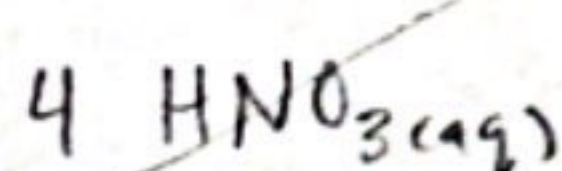
$$\Delta H_2 = -76.6 kJ$$



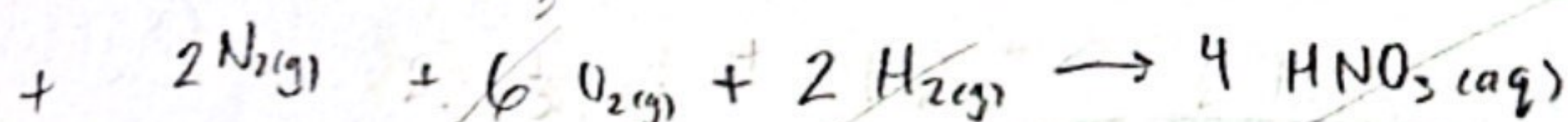
$$\Delta H_3 = -174.1 kJ$$



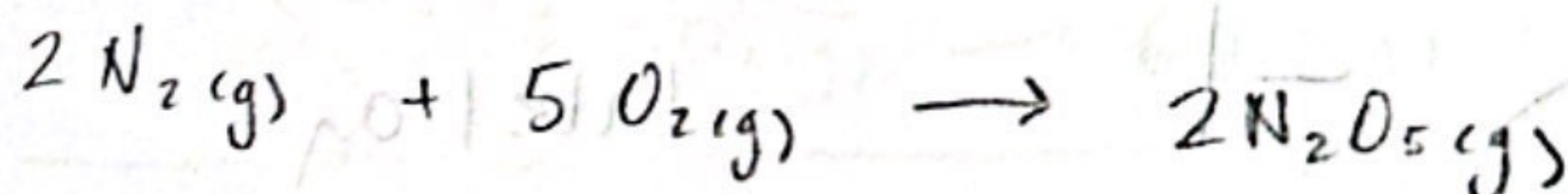
$$\Delta H_1 = -2(-285.8 kJ) = 571.6$$



$$\Delta H_2 = -2(-76.6 kJ) = 153.2$$

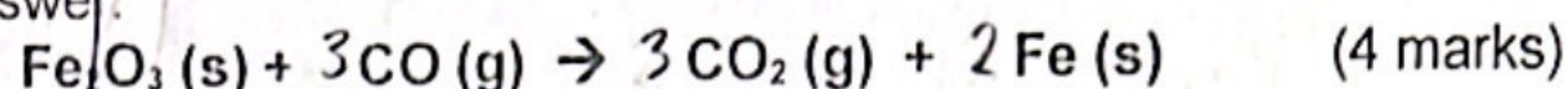


$$\Delta H_3 = 4(-174.1 kJ) = -696.4$$



$$\Delta H_1 + \Delta H_2 + \Delta H_3 = 571.6 kJ + 153.2 kJ - 696.4 kJ = 28.4 kJ$$

3. Determine the enthalpy change of the following reaction, using the Standard Molar Enthalpies of Formation listed on the Reference Tables provided. Make sure to balance the reaction with whole number coefficients and include an equation in your answer.



$$\Delta H^\circ = \sum n_{\text{products}} \Delta H^\circ_f \text{products} - \sum n_{\text{reactants}} \Delta H^\circ_f \text{reactants}$$

$$= [3\text{mol}(-393.5 \text{ kJ/mol}) + 2\text{mol}(0 \text{ kJ/mol})] - [1\text{mol}(-824.2 \text{ kJ/mol}) + 3\text{mol}(-110.5 \text{ kJ/mol})]$$

$$\Delta H^\circ = -1180.5 \text{ kJ} + 0 \text{ kJ} - (-824.2 \text{ kJ}) - (-331.5 \text{ kJ})$$

$$\Delta H^\circ = -24.8 \text{ kJ} - 0.5$$

$$\Delta H^\circ_f = \frac{\Delta H^\circ}{n} = \frac{-24.8 \text{ kJ}}{1 \text{ mol}} = -24.8 \text{ kJ/mol}$$

Application (6 Marks)

1. Identify the "system" and "surroundings" in the example. State whether the reaction is endothermic or exothermic. (2 marks): A snowflake lands on your hand

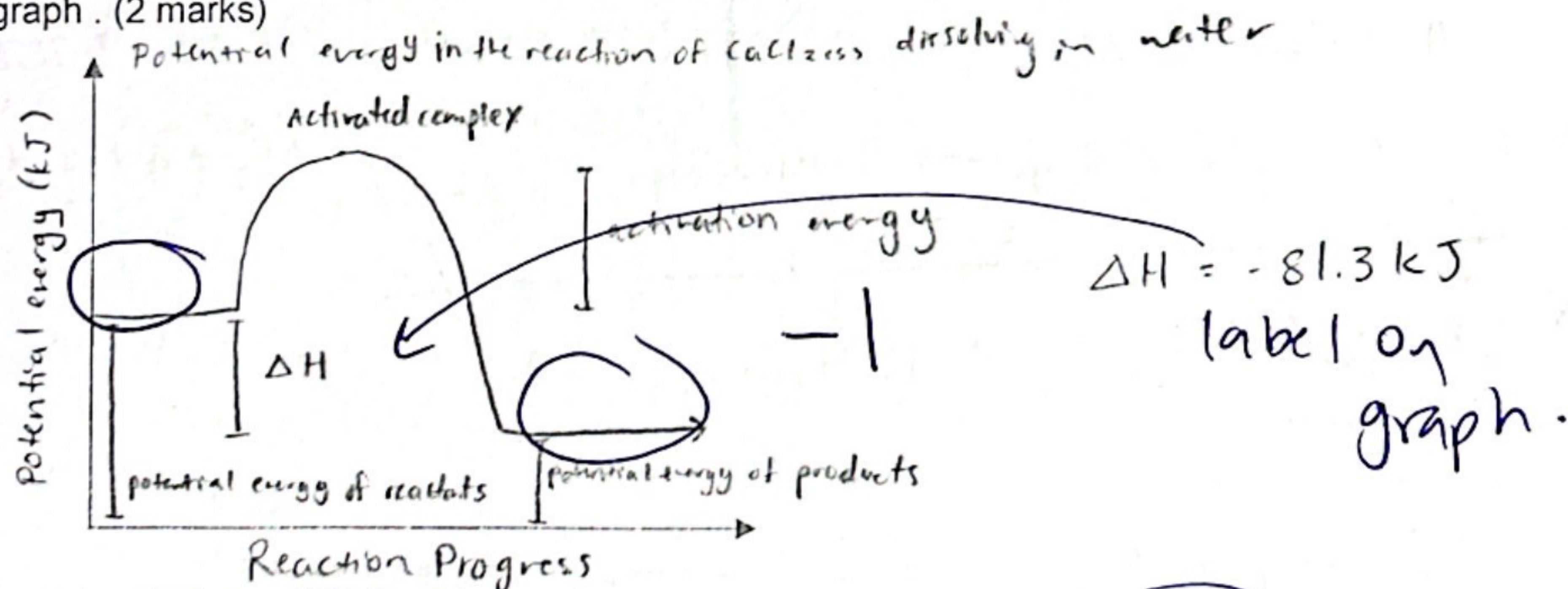
System: the water crystals that compose the snowflake — 0.5 $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$
 surroundings: your hand — 0.5 melt.

The heat from your hand is absorbed by the water, making the reaction endothermic (when it melts)

2. A hot pack consists of an inner pouch containing solid calcium chloride, $\text{CaCl}_2(\text{s})$, and an outer pouch of water. Twisting the pack breaks the inner pouch and allows the water and calcium chloride to mix. As the calcium chloride **dissolves**, the temperature of the surrounding environment increases. The amount of energy released by the system per mol of calcium chloride dissociated is 81.3 kJ.

a) Classify the reaction as endothermic or exothermic. exothermic (0.5 mark)

b) Sketch a potential energy diagram for the reaction. Label the change in enthalpy, ΔH and any other necessary parts on the graph. (2 marks)



c) Write a thermochemical equation for this reaction. (1.5 marks)

